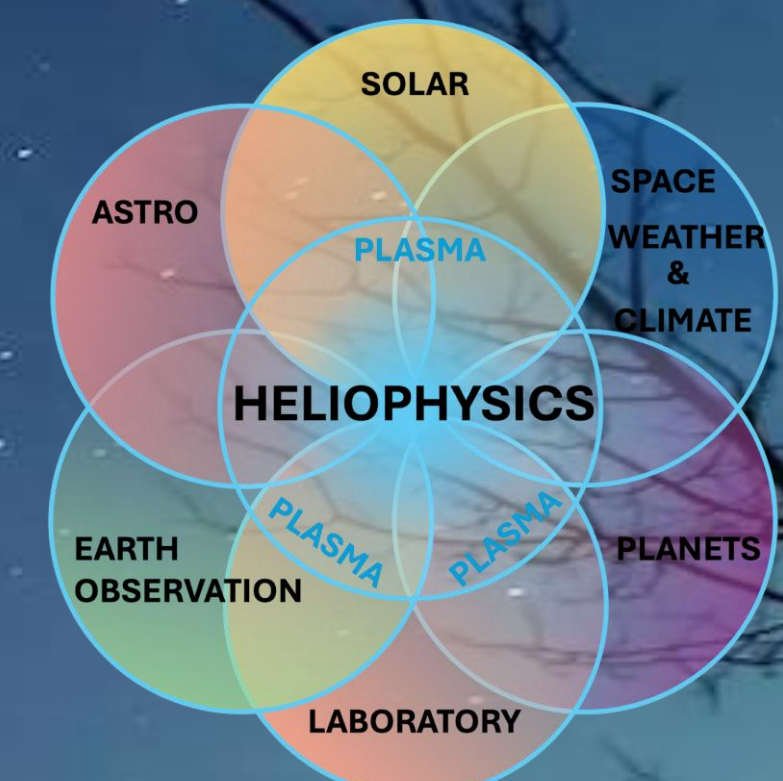




Comparison between Geomagnetic Storms of Different Intensities and their Correlation with Magnetospheric Substorms

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Introduction

Geomagnetic storms are phenomena that significantly disturb the magnetosphere of the Earth. **Magnetospheric substorms** are explosive events that take place in the magnetosphere-ionosphere system and can occur both during magnetic storms and during non-storm intervals. However, there is no strong statistical evidence supporting a causal relationship in which substorms trigger geomagnetic storms or vice-versa. In this study we compare two categories of magnetic storms, “**medium**” storms and “**intense**” storms with $-100 \text{ nT} \leq \text{Dst}_{\min} \leq -50 \text{ nT}$ and $\text{Dst}_{\min} \leq -130 \text{ nT}$, respectively. We identify both the count as well as the intensity of the **storm-time substorms** that take place during each storm, using the SML and AL indices. Our goal is to compare the performance of these indices that, although they track the same phenomenon, vary in their geomagnetic latitude distribution.

Datasets : Dst, AL, AU index data, provided from the WDC for Geomagnetism, Kyoto, and for the SML and SMU index data, we gratefully acknowledge the SuperMAG collaborators.

Analysis

- We use data for **100 geomagnetic storms** occurring between 1990 and 2015, 50 of which are classified as “medium” and 50 as “intense”.
- We extract the time series from -24 hours to +48 hours, in respect to the **$t=0$ of storm minimum**. For the same timespan, we extract the timeseries for the SML and AL indices for each geomagnetic storm.
- We identify magnetospheric substorms using a **modified** version of the **technique** from **Newell & Gjerloev**.
- Superposed epoch analysis** was applied to each of the **four dataset combinations**, medium or intense storms, each analyzed using either the SML/SMU or the AL/AU index pair.
- The identified substorms were binned in 6-hour time bins, for which the **mean intensity ΔSML or ΔAL**

Results

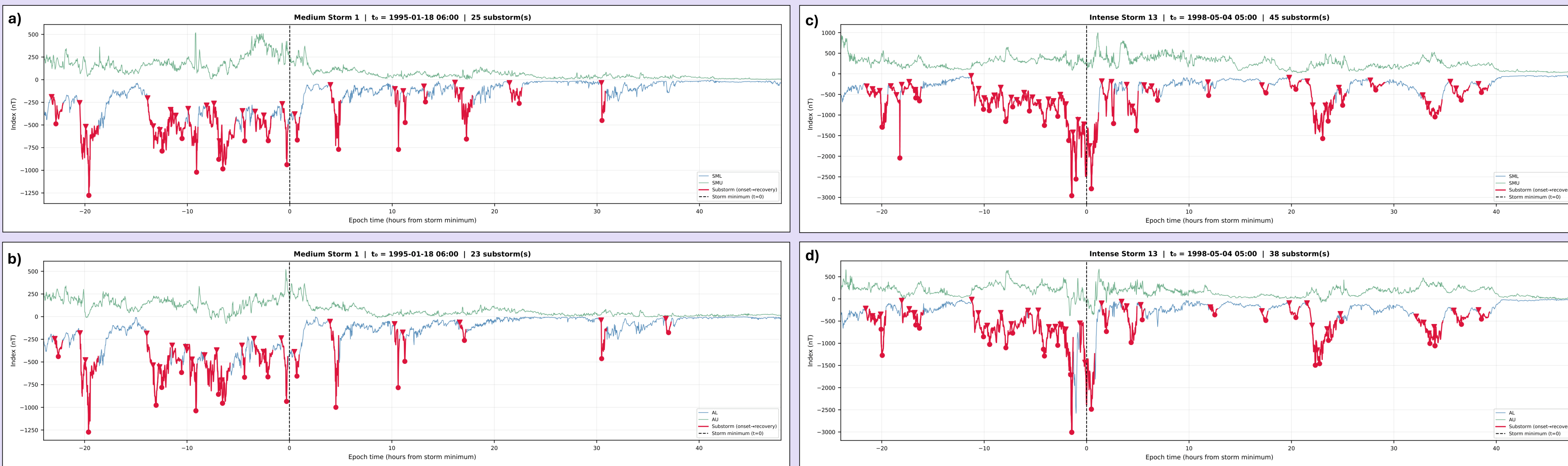


Figure 1: Substorm identification plots for two individual storm time series. Each substorm onset is marked by an inverted red triangle, while the corresponding substorm minimum is indicated by a red circle.

- Substorms in “Medium Storm 1” are identified using SuperMAG’s SML and SMU indices.
- Substorms in “Medium Storm 1” are identified using WDC’s AL and AU indices.
- Substorms in “Intense Storm 13” are identified using SuperMAG’s SML and SMU indices.
- Substorms in “Intense Storm 13” are identified using WDC’s AL and AU indices.

We can see that for the same storm timeseries, SML and AL identify approximately the same number of substorms, with SML identifying 8.7% and 18.4% more substorms for “Medium Storm 1” and “Intense Storm 13”, respectively. We also observe the presence of some intense substorms many hours after the storm minimum has passed, which is consistent with the general trend according to the following plots.

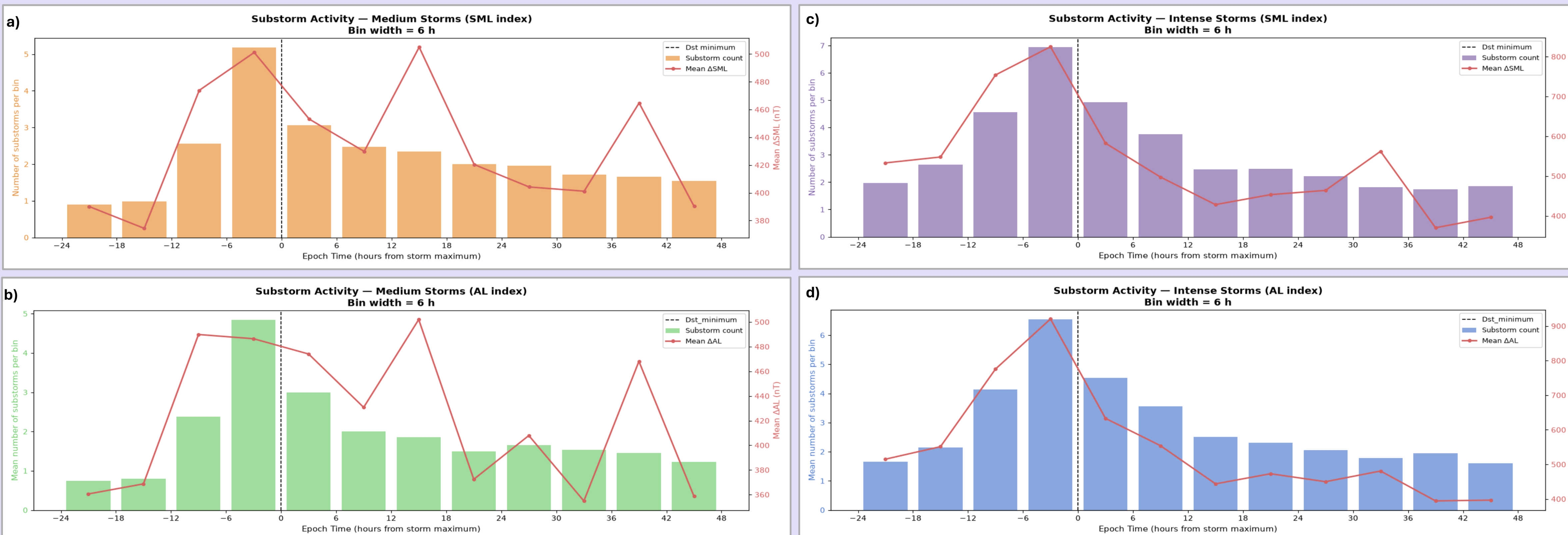


Figure 2: Superposed epoch analysis (SEA) of the substorms identified in all the storms, with respect to storm minimum.

- SEA of substorms identified using SuperMAG’s network, during Medium Storms.
- SEA of substorms identified using WDC’s network, during Medium Storms.
- SEA of substorms identified using SuperMAG’s network, during Intense Storms.
- SEA of substorms identified using WDC’s network, during Intense Storms.

The filled bars show the mean number of substorms in each 6-hour time bin. The red circles show the average value of ΔSML or ΔAL .

The trend that we observe for the intense storms is as expected for both indices. The substorm count as well as their mean intensity increase 12 hours prior to storm minimum, a timespan that includes the main phase of a storm. Afterwards, both the count and the intensity show a gradual decrease, with the exception of a peak in the [30, 36] hours bin which is more prominent in the SML analysis. As for the medium storms, we also observe an increase in both count and mean intensity of the substorms 12 hours prior to storm minimum. However, after the storm minimum, although the substorm count decreases, the mean intensity exhibits fluctuations that can be observed in both the SML and AL diagrams.

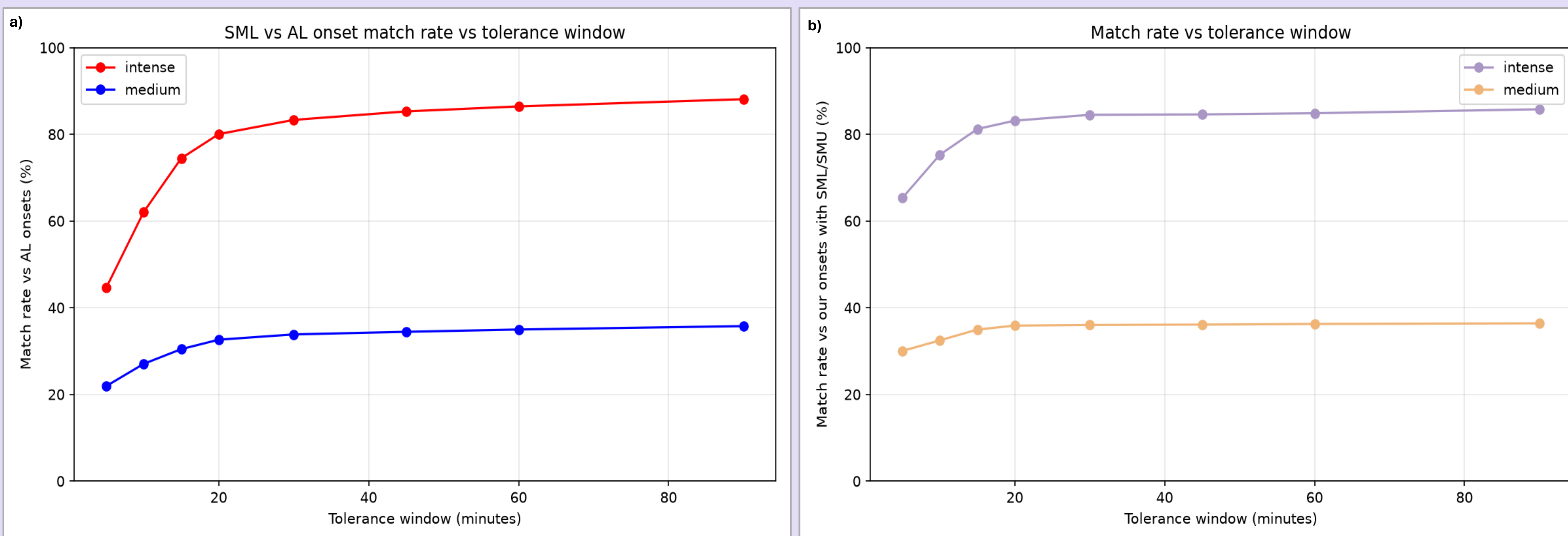


Figure 3: Substorm match rate (%) per tolerance window.

- Substorm match rate for substorms identified using SuperMAG’s network, compared with those identified using WDC’s network. The red line corresponds to substorms during Intense Storms and the blue line to those during Medium Storms.
- Substorm match rate for the substorms we identified using SuperMAG’s network, compared with those from Newell & Gjerloev’s list –also identified using SuperMAG’s network. The purple line corresponds to substorms during Intense Storms and the orange line to those during Medium Storms.

In figure 3a, we observe that the intense storm substorms category (red curve) exhibits a faster and more abrupt match rate, in contrast with the medium storm substorm category (blue curve), which shows a lower match rate in general. In figure 3b, we observe that the intense storm substorms category (purple curve) shows a considerably higher and faster-saturating match rate, in contrast with the medium storm substorm category (orange curve), which shows a substantially lower match rate throughout. It should be noted that the only criterion for this comparison was the onset datetime of each substorm. While this was the only possible approach, it is admittedly a fairly limited basis for comparison as it captures only timing agreement.

Main Conclusions

- It generally appears that the **greatest substorm activity**, both in terms of count and mean intensity, occurs in the **12-hour period preceding the storm minimum up to the storm minimum itself**. This pattern is consistent across both substorm detection indices as well as both storm types examined.
- For **Medium Geomagnetic Storms**, the SML index detects **8.3% more substorms** than AL within the 12-hour window prior to storm minimum, while **the average intensity** remains **nearly identical** between the two indices.
- For **Intense Geomagnetic Storms**, the SML index detects **7.4% more substorms** than AL in the same 12-hour window. However, substorms identified by the **AL** index show a **7% higher average intensity** compared to those identified by SML.
- In total, the SML index identified **1,319** and **1,868 substorms** during the **50 Medium** and **50 Intense Magnetic Storms** analyzed, respectively, compared to **1,150** and **1,738 substorms** identified by the **AL** index for the same events.
- Furthermore, the **stronger convection activity** associated with **Intense Magnetic Storms** may contribute to this effect, as enhanced convection can drive sustained current systems that the two indices resolve differently.
- As we see in Figures 3a and 3b for substorms during **medium** geomagnetic storms the **match rates** are considerably **lower**, saturating at **~35%**, which is not surprising, considering the discrepancies we observed between Figures 2a and 2b, which were more apparent while trying to match each specific substorm onset between the two data sets.
- Overall, the **comparison** between the AL/AU and SML/SMU indices has shown that the **reliability** and mutual agreement of substorms detected with data from **terrestrial magnetometers** is itself **intensity-dependent**, converging most closely during the most active periods of the most intense storms.

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